



Indian Institute of Technology Gandhinagar

Proposal for a New Course

Course No.:	EH 304								
Course Title:	Drone data acquisition, processing and interpretation								
Credits:	<table><tr><td>L</td><td>T</td><td>P</td><td>C</td></tr><tr><td>1</td><td>0</td><td>2</td><td>2</td></tr></table>	L	T	P	C	1	0	2	2
L	T	P	C						
1	0	2	2						
Prerequisites (if any):	None								
Potential Instructors:	Pankaj Khanna								
Expected frequency of offering this course	Each year								

Course contents:

DGCA guidelines on drone types as well as the laws and regulations for flying drones in India. Drone data acquisition methods and techniques, with nadir and oblique data acquisition and automated and manual data collection methods. Image processing using software’s (such as Agisoft and/or Pix4D), with an introduction to structure-from-motion methodology and development of 3D Digital Outcrop Models (DOM’s). Introduction to lidar, thermal, multi and hyperspectral drone datasets. Data interpretation using open-source software’s (such as cloud compare).

Lab

1. Demonstration of different drones and the different equipment’s/sensors.
2. Ground Control points with RTK within IIT GN campus.
3. Drone data acquisition within IIT GN campus.
4. Data processing with Agisoft: 3D Model building.
5. Drone data interpretation utilizing open-source software’s (such as Cloud Compare)

Texts/References:

- Frazier and Singh, 2021. Fundamentals of Capturing and Processing Drone Imagery and Data. CRC Press.
- Indian Government platform for drone related information: Digital Sky Platform - <https://digitalsky.dgca.gov.in/>

Learning outcome

- Have understanding of drone types, acquisition methods, and processing
- & interpretation software’s

- Will learn about the laws and regulations of flying drones in India
- Will get hands on experience on flying a drone

Any other Remarks

The course shall have field visits and laboratory demonstrations to reinforce the classroom teachings